

SMART DETECTION SYSTEM FOR FRUIT RIPENESS

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Abstract-With the increasing growth of world population, the agricultural industry is demanding technological solutions focused on growth of food production and farming to enhance the farm yield. Nowadays, technology is constantly improving to solve human problems more easily. The manual experimentation and assessment through trial and error etc are no longer practicable. The intelligent system can be established for fruits that can permit the quality analysis. The proposed system will provide a much better and convenient way for farmers to get the maturity of fruit. This includes a series of steps from capturing the image of fruit to identify whether the fruit is fully ripened or moderately ripened or immature through the implementation in Raspberry Pi. The camera module is interfaced with Raspberry pi and the captured images are examined by various steps like preprocessing and segmentation. The status of the fruit has been sent by GSM module through SMS and displaying the status on the monitor display of the owner of the system. This in turn reduces the need for labor in fruit farms. This smart way of analyzing the fruit will enable growers and farmers and helps them to understand the potential of IoT market for agriculture by installing smart technologies to increase competitions and sustainability in their production. This will further help for the automatic fruit harvesting system using robotic arm.

Keywords- Fruit Ripeness, IoT, Foldscope, MATLAB, Raspberry pi, OpenCV- Python, GSM.

I. INTRODUCTION

Plants have become a primary source of energy and to solve the global warming problem. Nowadays, automation plays an major role in various fields. Majority population in India dependent on agriculture and becomes their fundamental point of income. In India, fresh fruits exportation is increased. Agriculture is considered as an art and because of science, agriculture is easier and efficient for farmer to attain greater output at the terminating stage of growing time [1].

As the competition is increased in international and domestic fruit markets, it is compulsory to improve ripeness evaluation techniques to decrease the loss occurred to the cultivator and packer from fruit spoilage.

Fruits are now available at both fresh and artificial processed forms. Processed forms include jam, paste and juice. The exportation of those processed foods yielding more income for the country.

People are very conscious about their health; they prefer only fresh, good quality fruit. In order to get good quality of processed products the quality of fruits should be good. Identifying good and bad quality fruits in industries manually is the main barrier since it is time consuming and requires high labor cost. Therefore high priority is given in fruit industries for various necessities to identify the quality of fruits.

The agricultural industry is demanding technological solutions focused on increasing production and benefits while reducing cost and time. Automation of agriculture tasks has improved all phases of the industrial process, from pre-harvest to harvest and post-harvesting stages, fruit plucking and collections of plucked fruits is the most important task.

The project describes an approach to image processing method in industries now become a major source in the last few years. By using Image processing tools in MATLAB software [2], we can find the quality of Fruits using various algorithms. During various ripening stage, the color of lemon differ from mature green, green and yellow. Most of them were based on color feature because the color is as an effective factor of the fruit quality. The fruit ripeness can be calculated by its surface color of fruit and it is a majority factor while finding a fruit ripeness.

The purpose of the present work is to analysis the age factor of the fruits using the technique mentioned that which will try to decrease the labor work and maintain the accuracy at its maximum, also the proposed technique is allow the process to become faster as compare to the manual system. It uses the various sub image processing algorithms to perform analysis.

It is very difficult task to categorize and ordering the fruits in a production farm. In order to solve this problem, MATLAB software and also python programming with some image processing methods are adopted widely. It is very necessary to maintain the proper quality of fruit for

maximum profit for growers while marketing the fruits. Hence the proposed model compacts with the valuation of ripeness using image processing methods.

The hardware verification about the ripeness state using Raspberry pi attached with a Raspberry pi camera that used to capture the image of the fruit and the color sensor is used to find the color level estimation of the fruit. The captured image undergoes various techniques and the status of the fruit which is fully ripened or moderately ripened or immature will be send to the user via SMS using GSM module. This system will be very convenient to find the ripeness state of the given fruit and instantly the user will receive the message of the fruit ripeness which makes the system much better to use.

II. PROPOSED DESIGN

Fig. 1 shows the Block Diagram of the Proposed System.

- The proposed system will take input as fruit and checks whether the fruit is fully ripened or moderately ripened or immature.
- The first system will take the picture of fruits and gives whether the fruit is in fully ripened or moderately ripened or immature state using MATLAB Image processing tools.
- The second system will be real time ripeness detection of fruit using Raspberry pi interfaced with camera module
- The third system requires the fruit sample as input and the samples is observed in foldscope to find out whether the fruit is immature or mature.

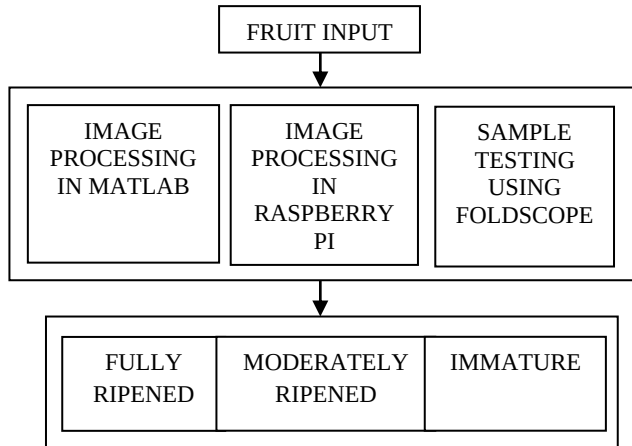


Fig.1 Block Diagram

2.1 ASSESSMENT OF MATURITY USING MATLAB

The proposed algorithm is implicit for the assessment of maturity. The RGB values of the image are retrieved. In this algorithm B value is considered to be zero based on the

RGB color coding of different shades of green and yellow color. The ratio between the R and G value is computed for a variety of sweet lime fruits [3]. The RG ratios for different samples were obtained. A region of interest (ROI) is a portion of an image that need to focus for further analysis. Here, three cases of maturity assessment is formulated,

1. Maturity analysis based on “Red &Green (R/G)” ratio.
2. Maturity analysis based on Hue Mean.
3. Maturity analysis based on Hue Median.

After Reading the image, the required part of the region is selected and the value of RGB is estimated.

2.1.1 MATURITY ANALYSIS BASED ON “Red &Green (R/G)” RATIO

Based on the findings, the RG ratio threshold for ripe fruit is fixed as greater than 0.95, and that for unripe fruit is less than 0.85, and between 0.85 and 0.95 is moderately ripened fruit. Fig. 2.1.1 shows the flowchart for assessment of fruit ripening based on (R/G) Ratio.

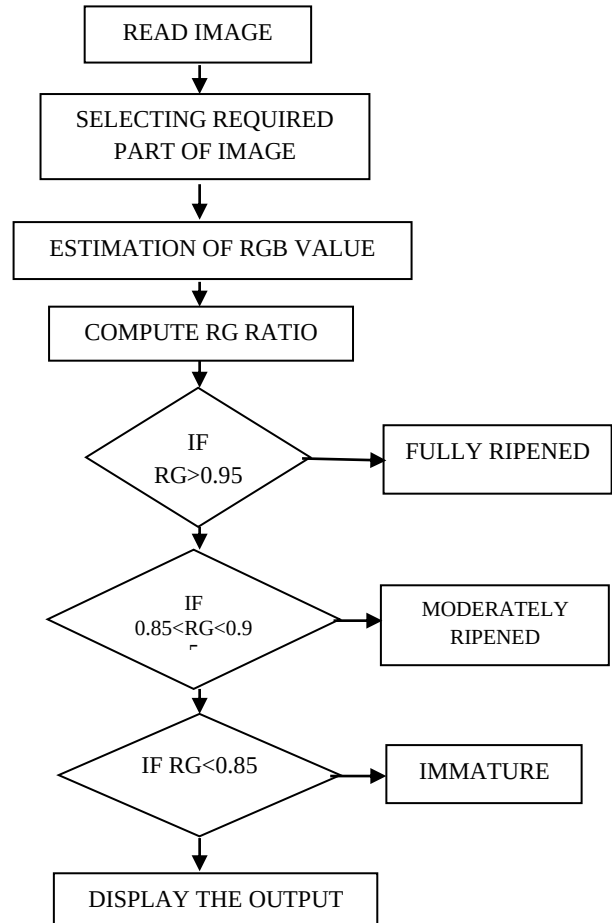


Fig 2.1.1: Flow chart based on R/G Ratio

2.1.2 MATURITY ANALYSIS BASED ON HUE MEAN

Convert the RGB to HSV color space. Calculate the Hue value of the selected region of image. The Hue Mean threshold for ripe fruit is fixed as less than 0.20, and that for unripe fruit is greater than 0.24 and between 0.20 and 0.24 is moderately ripened fruit. Fig. 2.1.2 shows the flowchart for assessment of fruit ripening based on Hue Mean.

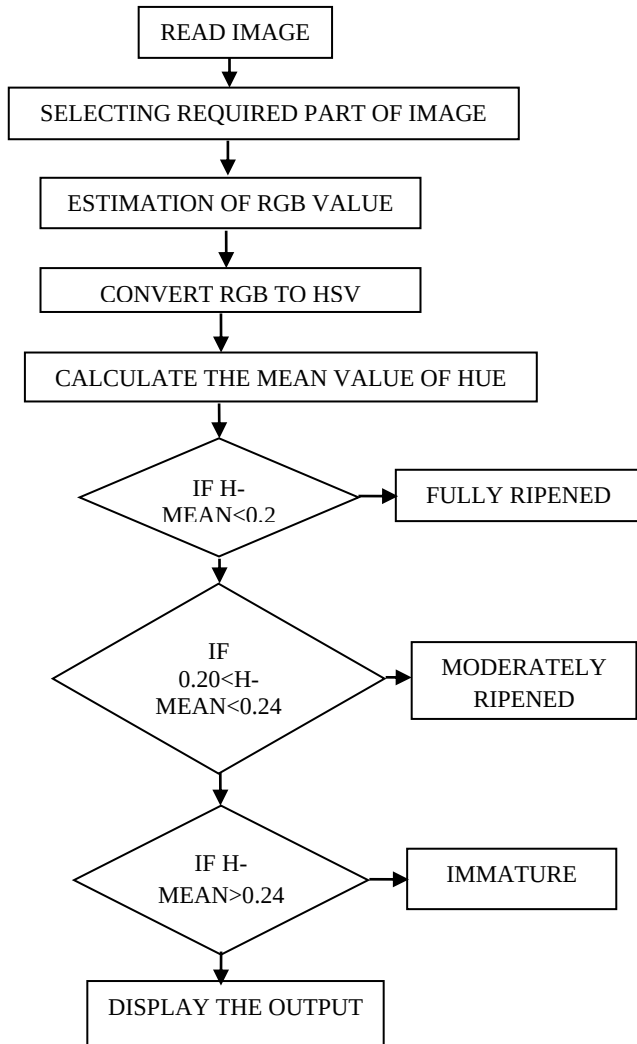


Fig 2.1.2: Flow chart based on Hue Mean.

2.1.3 MATURITY ANALYSIS BASED ON HUE MEDIAN

Convert the RGB to HSV color space. Calculate the Hue value of the selected region of image. The Hue Median threshold for ripe fruit is fixed as less than 0.19, and that for unripe fruit is greater than 0.22 and between 0.19 and 0.22 is moderately ripened fruit. Fig. 2.1.3 shows the

flowchart for assessment of fruit ripening based on Hue Median

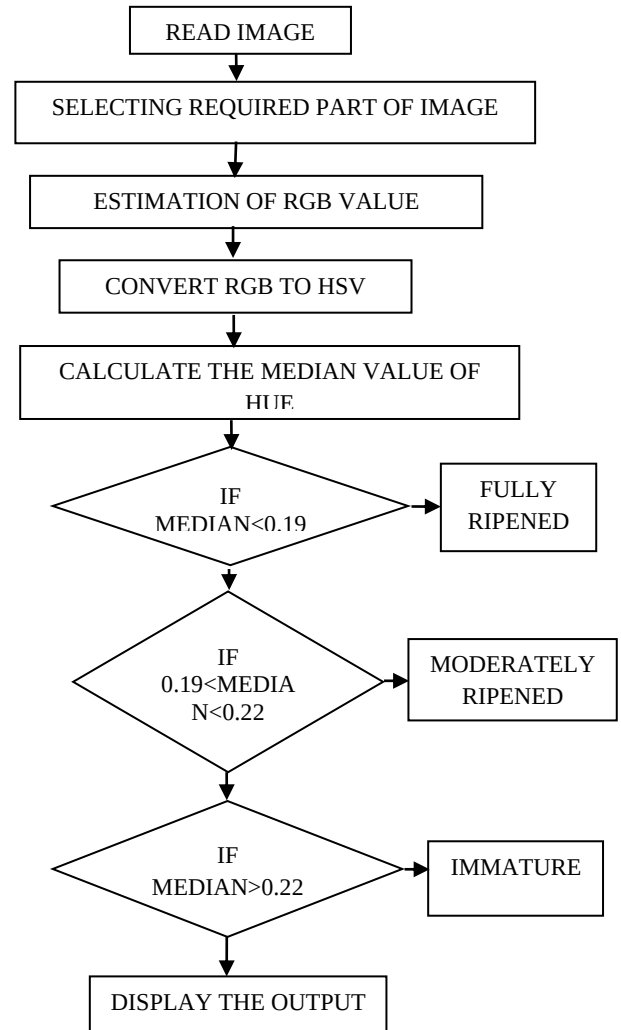


Fig 2.1.3: Flow chart based on Hue Median.

2.2 ASSESSMENT OF MATURITY USING RASPBERRY PI

A powerful small single board computer called Raspberry Pi using Raspbian as its Operating system which is a version of the GNU/Linux. The processor of Raspberry Pi system is a Broadcom BCM2835 system-on-chip (SoC) multimedia processor, consist of a Videocore4 GPU and 64 bit Advanced RISC Machine (ARM) CPU core, specifically ARM Cortex - A53 and it runs at 1.2 GHz with RAM of 1 GB.

An open library that has more number of algorithms in computer vision is OpenCV (Open Source Computer Vision Library). The OpenCV allows for automatic memory allocation and memory de-allocation

and reduces the total programming complexities for programmers. OpenCV-Python is the OpenCV integrated with the programming language Python. The syntax of python is simple and easy to use. OpenCV-Python provides common functions for computer vision applications and to increase the use of machine approach in the commercial products. OpenCV-Python focuses mainly on real-time image processing [4].

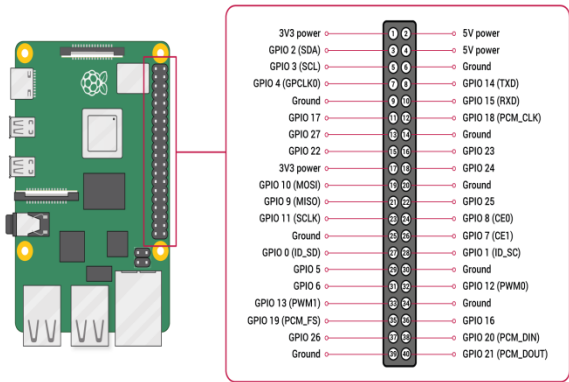


Fig.2.2.1. Raspberry Pi 3 B



Fig.2.2.2 Raspberry Pi camera

Fig. 2.2.2 shows the camera module for Raspberry Pi, it has the resolution of 5MP.



Fig.2.2.3

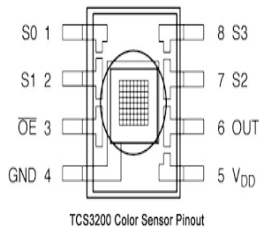


Fig.2.2.4

Fig.2.2.3 shows the Color Sensor TCS3200 and Fig. 2.2.4 shows the Pinout of TCS3200

Figure (2.2.3) shows the color Sensor TCS3200 which exactly a color detector. It consists of four white LEDs and a RGB sensor chip of TAOS TCS3200. It will detect visible colors and measure colors within limitless range to a certain degree. This sensor has filter types which are: red filter, green filter, blue filter and clear with no filter. With the high and low value of pin S2 and S3 on the module the filter is selected. The GSM mobile telephone technology is the GSM module which is the hardware device, the remote network will get the data link using GSM. Figure (2.2.5) shows the GSM module SIM900A which is designed with a very powerful single-chip processor integrating AMR926EJ-S core.

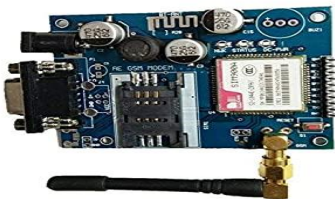


Figure 2.2.5: GSM module SIM900A

2.2.1 FLOW CHART

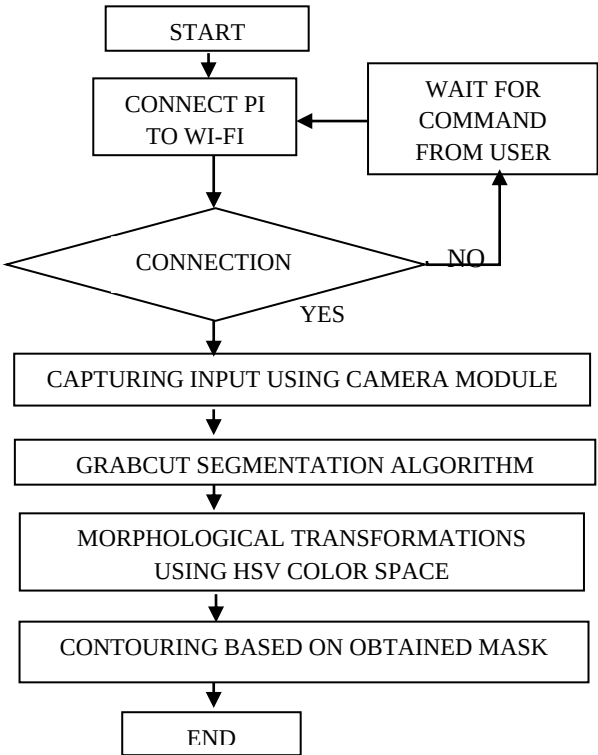


Fig 2.2.6: Flow chart for Raspberry Pi

The working of flowchart is first the pi connects to the Wi-Fi connectivity. If the connection is successful, it connects to Wi-Fi, and then camera module captures the input fruit. The captured image will undergo image segmentation. For this Grabcut image segmentation algorithm will segment the image background and the final segmented image undergoes morphological transformations using HSV color model. After that it contours based on obtained mask. The final result shows that given fruit is fully ripened or moderately ripened or immature. Fig.2.2.6 shows the flow of process for Raspberry Pi fruit ripeness detection technique.

2.2.2 FLOW CHART OF COLOR SENSOR

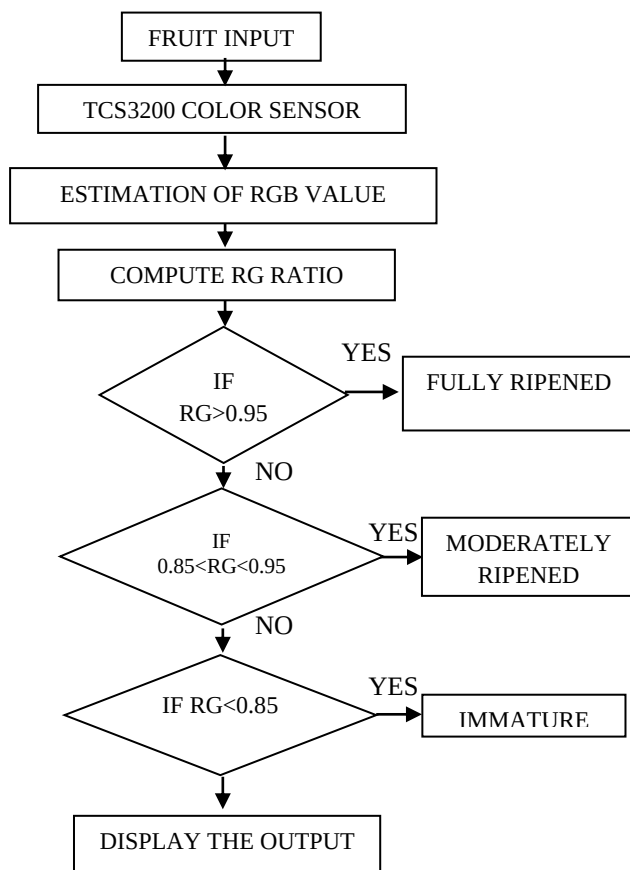


Fig 2.2.7: Flow chart for Color sensor

The flow chart for maturity assessment using Color sensor was shown in Figure 2.2.7. After placing the fruit input the color sensor which senses the color of the fruit. The RGB values of the image are retrieved. The ratio between the R and G value is computed for a variety of sweet lime fruits. The RG ratios for different samples were obtained. With the obtained findings, the RG ratio threshold for ripe fruit is fixed as greater than 0.95, and that for the unripe fruit is less than 0.85, and between 0.85 and 0.95 is moderately ripened fruit.

2.3 ASSESSMENT OF MATURITY USING FOLDSCOPE

A portable optical microscope and it will be assembled using a paper sheet and lens called Foldscope. The foldscope cost is very cheap and used to observe biological and a non-biological science. Foldscope is water-proof when compared to conventional microscope [5]. The biological samples such as fungi, bacteria, etc and also non-biological samples will be visualized using foldscope. The assembled foldscope with various parts is illustrated in Fig.2.3.1

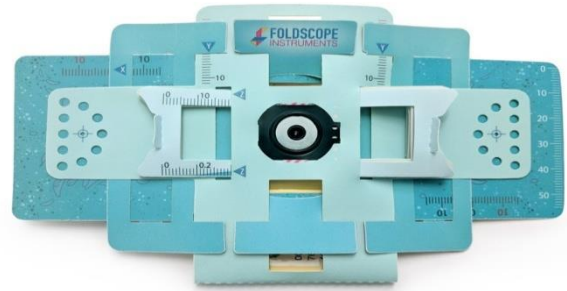


Fig.2.3.1. Foldscope

A foldscope includes a cardstock having punched sheet, a spherical lens and three magnetic couplers which is square in shape. The lens having the magnification of 140X and 2 microns resolution is used in foldscope. The Weight of Foldscope was 8 grams. Foldscope will be effortlessly attached with mobile phone to get images for further analysis. The foldscope attached with the mobile phone is illustrated in Fig.2.3.2



Fig.2.3.2. Foldscope attached with mobile phone

III. RESULTS AND DISCUSSION

3.1 MATLAB RESULTS

The lemon images are taken which is green to yellow shades. After that selecting the required part of the image and simulated using three algorithms which are maturity analysis based on "Red & Green (R/G)" ratio; Hue Mean and Hue Median. The Algorithms are simulated by

MATLAB R2018a version. The outputs of these three algorithms are displayed in Command window of MATLAB whether fully ripened or moderately ripened or immature.

3.1.1 FULLY RIPENED SAMPLE



A Input image B Selecting Region C Required Image

```
Command Window
New to MATLAB? See resources for Getting Started.

>>
>> review3
(1) FULLY RIPENED; (2) FULLY RIPENED; (3) FULLY RIPENED; >> ;
f3 |
```

D Output Display

Fig.3.1.1. Fully ripened sample

Fully Ripened Lemon images are taken which as illustrated in the Fig.3.1.1 A Input image; selecting the region required as illustrated in Fig. 3.1.1 B Selecting Region; selected image as illustrated in Fig.3.1.1 C Required image ; output of three algorithms are displayed as illustrated in Fig.3.1.1. D output display

3.1.2 MODERATELY RIPENED SAMPLE



A Input image B Selecting Region C Required Image

```
Command Window
New to MATLAB? See resources for Getting Started.

>> review3
(1) MODERATELY RIPENED; (2) MODERATELY RIPENED; (3) MODERATELY RIPENED; >>
>>
>>
>>
```

D Output Display

Fig.3.1.2. Moderately ripened sample

Moderately Ripened Lemon images are taken which as illustrated in the Fig.3.1.2 A Input image; selecting the region required as illustrated in Fig. 3.1.2 B Selecting Region; selected image as illustrated in Fig.3.1.2 C Required image ; output of three algorithms are displayed as illustrated in Fig.3.1.2. D output display

3.1.3 IMMATURE SAMPLE



A Input image B Selecting Region C Required Image

```
Command Window
New to MATLAB? See resources for Getting Started.

>>
>>
>>
>>
>>
```

D Output Display

Fig.3.1.3. Immature sample

Immature Lemon images are taken which as illustrated in the Fig.3.1.3 A Input image; selecting the region required as illustrated in Fig. 3.1.3 B Selecting Region; selected image as illustrated in Fig.3.1.3 C Required image ; output of three algorithms are displayed as illustrated in Fig.3.1.3. D output display

3.2 RASPBERRY PI RESULTS

The Camera module and color sensor is connected to Raspberry pi. Raspberry pi connects to the Wi-Fi connectivity, if the connection is successful, it connects to Wi-Fi, then camera module captures the input fruit and it undergoes Grabcut segmentation and morphological transformations using HSV color space. After that it contours based on obtained mask. The final result shows the fully ripened region or moderately ripened or immature region via GSM module to user.

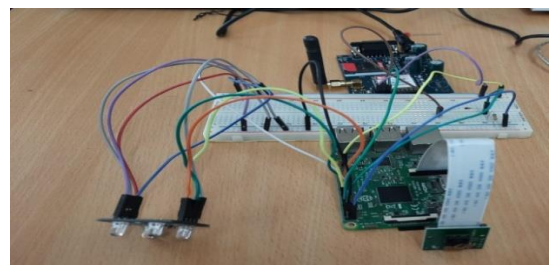


Fig.3.2.1. Hardware setup of Raspberry Pi

3.2.1 FULLY RIPENED SAMPLE

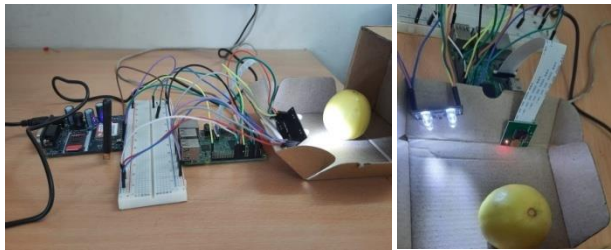


Fig. 3.2.1.1 A Fig. 3.2.1.1 B

Fig. 3.2.1.1: Fully Ripened fruit sample
(3.2.1.1 A) Side View of hardware setup, (3.2.1.1 B) Top View of sample

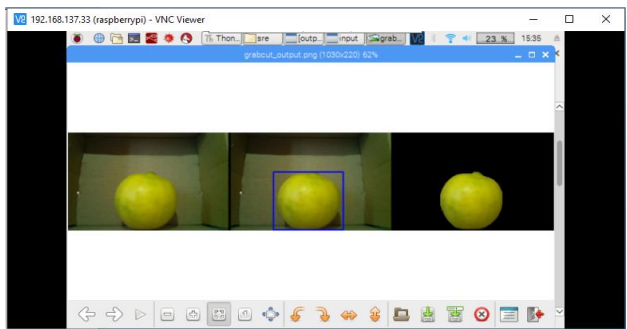


Fig.3.2.1.2 (a) Captured Fruit, (b) Rectangular region for grabcut segmentation, (c) After segmentation.

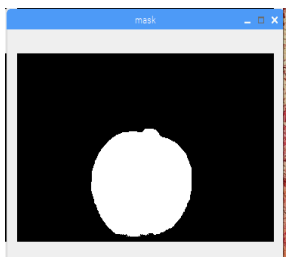


Fig. 3.2.1.3

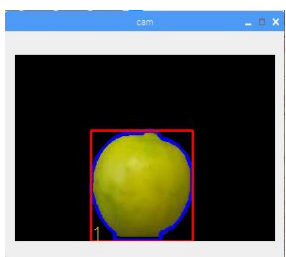


Fig. 3.2.1.4

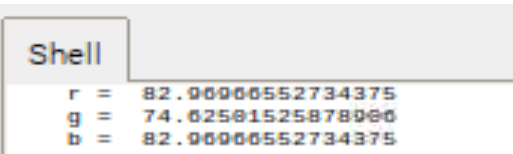


Fig. 3.2.1.5



Fig. 3.2.1.6

Fig.3.2.1.3 Morphological transformation of HSV Color masked image, Fig.3.2.1.4 Output Contoured image,

Fig.3.2.1.5 Color Sensor Result, Fig.3.2.1.6 Output Display.

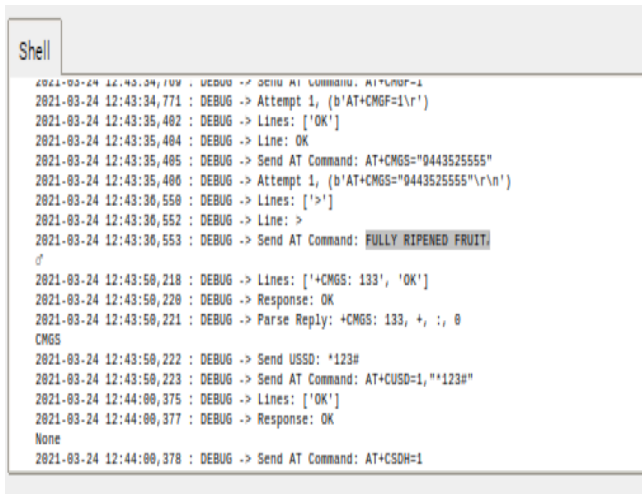


Fig. 3.2.1.7 GSM Monitor Display

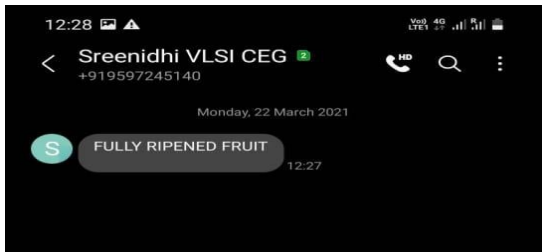


Fig. 3.2.1.8 Mobile Phone Result

Figure 3.2.1.1 A shows the full hardware setup for fruit ripening, Figure 3.2.1.1 B clearly shows the top view of the setup. In that Fully ripened Lemon fruit placed in front of camera module and the color sensor. Figure 3.2.1.2.a shows the captured fruit image and Figure 3.2.1.2.b shows the Rectangular region for Grabcut image segmentation algorithm. Figure 3.2.1.3 shows the Morphological operation of the segmented image using HSV color mask for green to yellow values and based on these color masking, contouring was done to detect the ripeness state and the contoured image is shown in Figure 3.2.1.4. Figure 3.2.1.5 shows the RGB value of the input image using color sensor. The final status of the fruit is shown in the figure 3.2.1.6. The Obtained result is transmitted to the user via GSM module and the Figure 3.2.1.7 Shows GSM module results and the Figure 3.2.1.8 Show the received message in mobile phone.

3.2.2 MODERATELY RIPENED SAMPLE

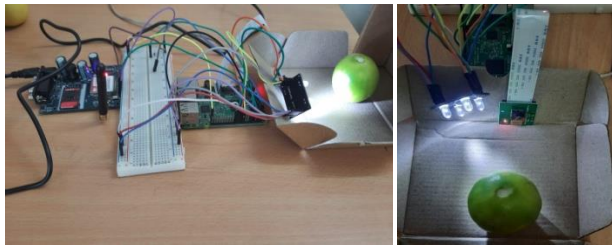


Fig.3.2.2.1 A Fig.3.2.2.1 B
Figure 3.2.2.1: Fully Ripened fruit sample
(3.2.2.1 A) Side View of hardware setup, (3.2.2.1 B) Top View of sample

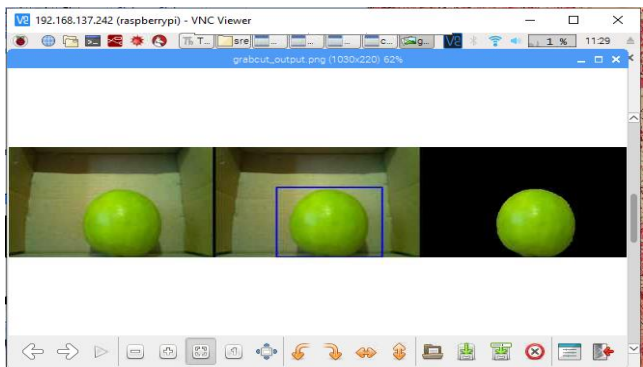


Fig.3.2.2.2 (a) Captured Fruit, (b) Rectangular region for grabcut segmentation, (c) After segmentation.

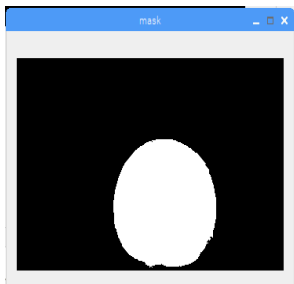


Fig.3.2.2.3

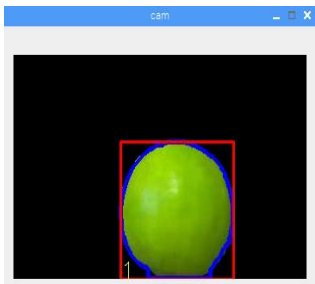


Fig.3.2.2.4

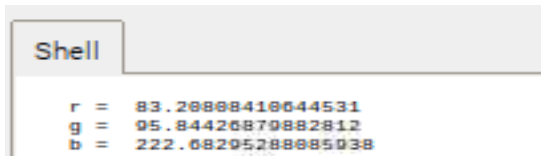


Fig. 3.2.2.5



Figure 3.2.2.6

Figure (3.2.2.3) Morphological transformation of HSV Color masked image, Figure (3.2.2.4) Output Contoured image, Figure (3.2.2.5) Color Sensor Result, Figure (3.2.2.6) Output Display.

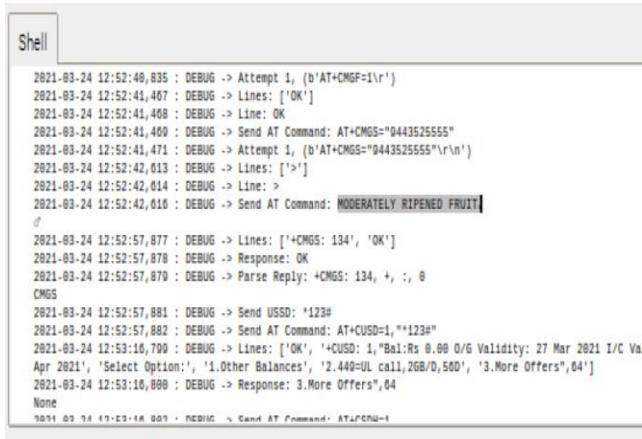


Fig.3.2.2.7 GSM Monitor Display

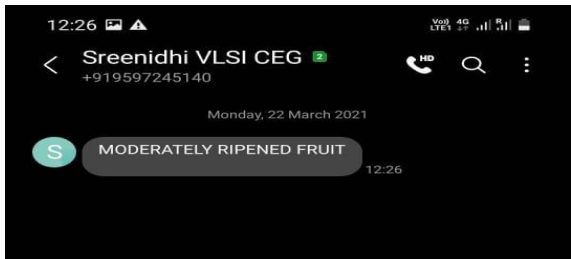


Fig.3.2.2.8 Mobile Phone Result

Figure 3.2.2.1 A shows the full hardware setup for fruit ripening, Figure 3.2.2.1 B clearly shows the top view of the setup. In that moderately ripened Lemon fruit placed in front of camera module and the color sensor. Figure 3.2.2.2.a shows the captured fruit image and Figure 3.2.2.2.b shows the Rectangular region for Grabcut image segmentation algorithm. Figure 3.2.2.3 shows the Morphological operation of the segmented image using HSV color mask for green to yellow values and based on these color masking, contouring was done to detect the ripeness state and the contoured image is shown in Figure 3.2.2.4. Figure 3.2.2.5 shows the RGB value of the input

image using color sensor. The final status of the fruit is shown in the figure 3.2.2.6. The Obtained result is transmitted to the user via GSM module and the Figure 3.2.2.7 Shows GSM module results and the Figure 3.2.2.8 Show the received message in mobile phone.

3.2.3 IMMATURE SAMPLE

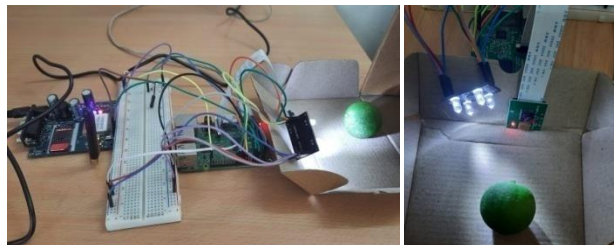


Fig.3.2.3.1 A

Fig.3.2.3.1 B

Fig. 3.2.3.1: Fully Ripened fruit sample

(3.2.3.1 A) Side View of hardware setup, (3.2.3.1 B) Top View of sample

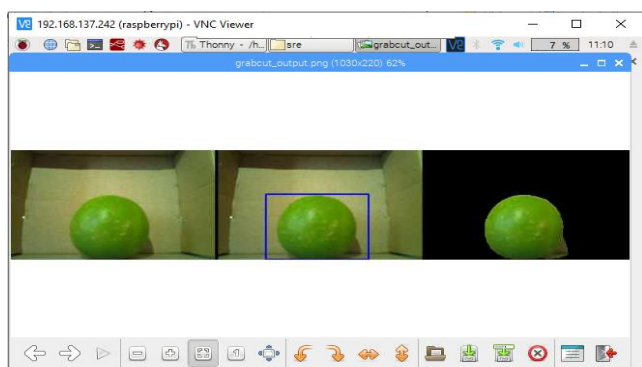


Figure 3.2.3.2 (a) Captured Fruit, (b) Rectangular region for grabcut segmentation, (c) After segmentation.

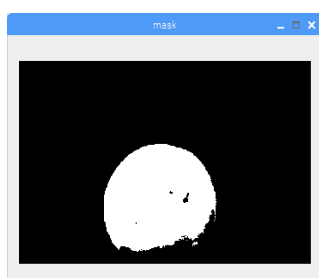


Fig.3.2.3.3

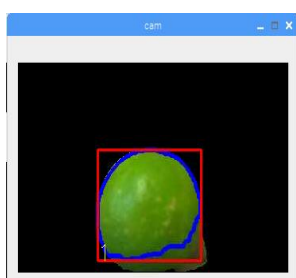


Fig.3.2.3.4

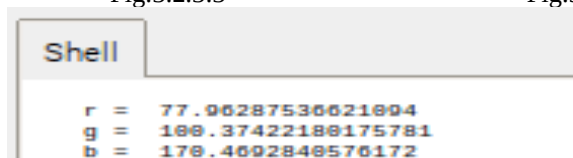


Fig.3.2.3.5



Fig.3.2.3.5

Figure (3.2.3.3) Morphological transformation of HSV Color masked image, Figure (3.2.3.4) Output Contoured image, Figure (3.2.3.5) Color Sensor Result, Figure (3.2.3.6) Output Display.

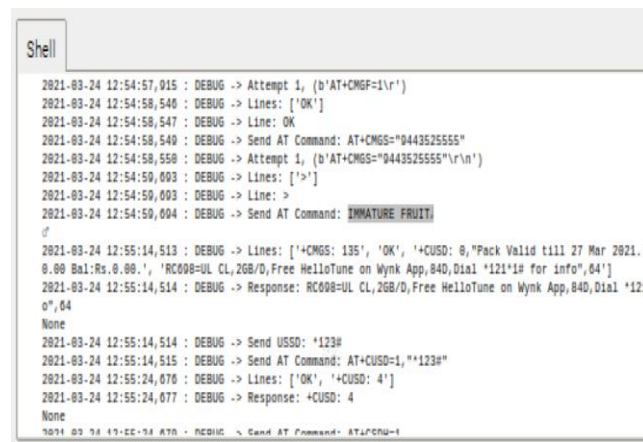


Fig. 3.2.3.7 GSM Monitor Display



Fig.3.2.3.8 Mobile Phone Result

Figure 3.2.3.1 A shows the full hardware setup for fruit ripening, Figure 3.2.3.1 B clearly shows the top view of the setup. In that Immature Lemon fruit placed in front of camera module and the color sensor. Figure 3.2.3.2.a shows the captured fruit image and Figure 3.2.3.2.b shows the Rectangular region for Grabcut image segmentation algorithm. Figure 3.2.3.3 shows the Morphological operation of the segmented image using HSV color mask for green to yellow values and based on these color masking, contouring was done to detect the ripeness state and the contoured image is shown in Figure 3.2.3.4. Figure 3.2.3.5 shows the RGB value of the input image using color sensor. The final status of the fruit is shown in the

figure 3.2.3.6. The Obtained result is transmitted to the user via GSM module and the Figure 3.2.3.7 Shows GSM module results and the Figure 3.2.3.8 Show the received message in mobile phone.

3.3 FOLDSCOPE OBSERVATIONS

The sample of fully ripened lemon and over ripened lemon was taken. The juice of those taken samples are viewed in foldscope and captured via mobile phone. The observed images are shown in Fig.3.3.1 which is fully ripened lemon sample and Fig. 3.3.2 which is over ripened lemon sample.

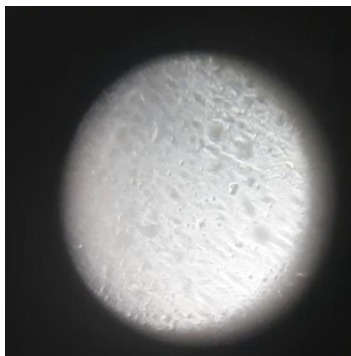


Fig.3.3.1 Fully Ripened



Fig.3.3.2 Over ripened

IV. CONCLUSION

Maturity of fruits acquired here is from fruit color. As color is the prime parameter for maturity evaluation, with the same variety of fruits and the color variations of the given fruit, color classification will find the fruit maturity. The project is mainly concentrating on reducing human effort and making human life easier. The first part of proposed system which is image processing will help to identify the ripeness especially citrus fruits, was implemented using MATLAB Image processing tools and second part was implemented by Raspberry pi and the third part is the verification internal quality by Foldscope. It's

our responsible to improve agricultural yielding. Hence, this project is purely focusing on Agricultural side. This concept of identifying the fruit ripening using Raspberry Pi will later helpful to build automatic plucking of Ripened Fruit using robotic arm.

V. FUTURE WORK

Future work that can be added to this project may be,

- To store the data into the cloud and communicate to the IoT device.

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